

QUANTLAB

US Equities Historical Replay

Architecture Options - Baseline v2

Primary market-data decision: Massive (formerly Polygon.io) US Stocks SIP

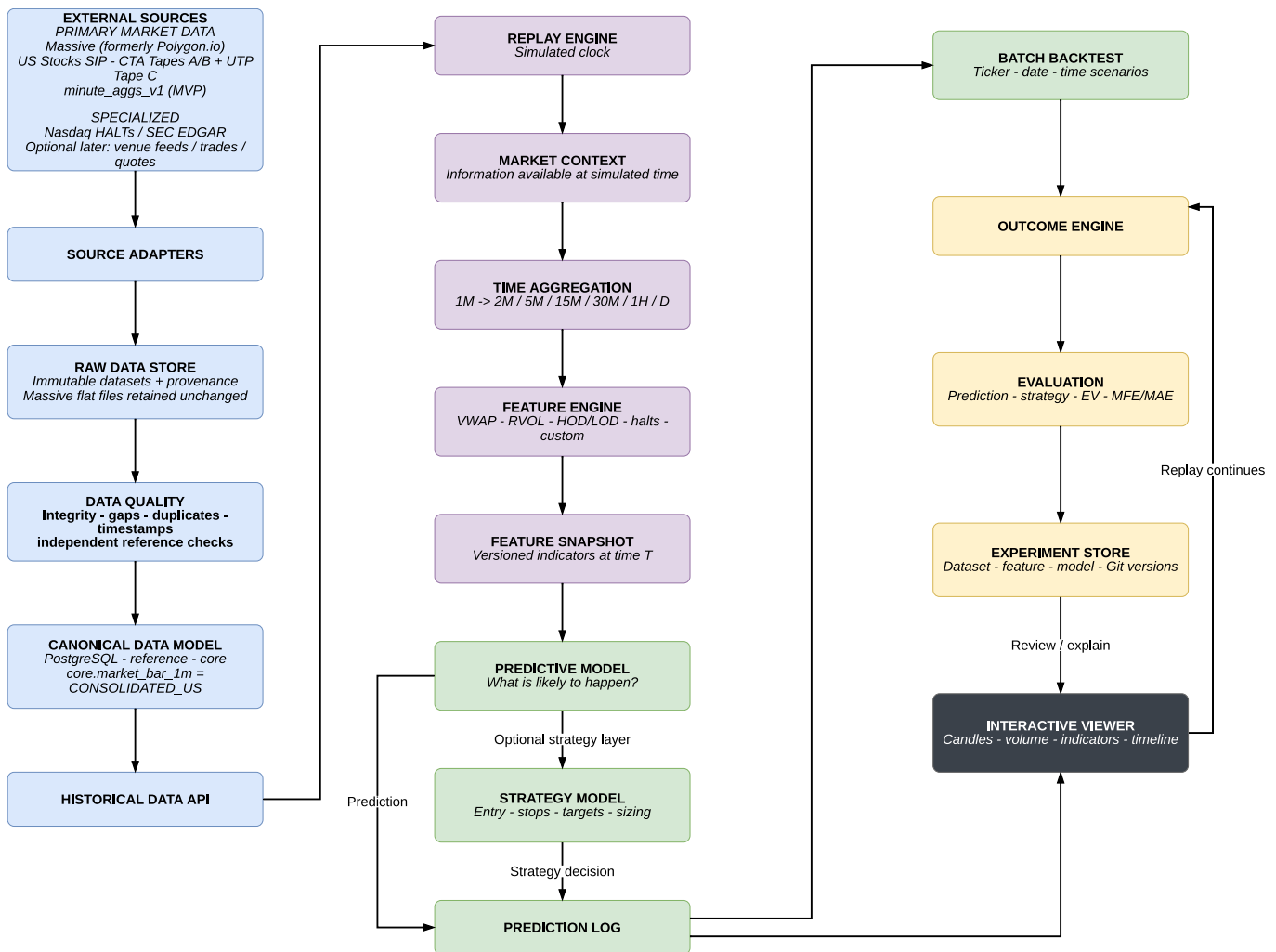
1. Architecture decision update

The architecture is updated to make Massive (formerly Polygon.io) the primary consolidated U.S. equities market-data provider for the MVP. Massive renamed Polygon.io on October 30, 2025; the provider states that its APIs, accounts and data products continued through the rebrand.

The MVP source is the US Stocks SIP dataset. The primary historical input is the daily minute aggregate flat file, `us_stocks_sip/minute_aggs_v1`, which provides OHLCV at one-minute granularity across U.S. equities. Massive documents that its stock flat files are consolidated and processed from regulated SIP data and include major U.S. exchanges, FINRA facilities and dark-pool reporting.

2. Functional reference diagram

The accompanying draw.io diagram preserves the structure of the supplied functional reference: Sources -> Adapters -> Raw Store -> Data Quality -> Canonical Data Model -> Historical Data API -> Replay Engine -> Market Context -> Time Aggregation -> Feature Engine -> Predictive/Strategy layers -> Batch/Viewer -> Outcome/Evaluation -> Experiment Store.



The source block is now expanded to explicitly show Massive/Polygon as the primary consolidated market source.

```

PRIMARY MARKET DATA
Massive (formerly Polygon.io)
US Stocks SIP
  - CTA Tapes A/B
  - UTP Tape C
  - minute_aggs_v1 = MVP source
  - trades_v1 / quotes_v1 = optional validation or future research
  
```

3. Data-source roles

Need	Selected source / role	MVP status
Consolidated U.S. OHLCV 1M	Massive US Stocks SIP - minute_aggs_v1	Primary
Consolidated U.S. trades	Massive US Stocks SIP - trades_v1	Optional validation / later
Top-of-book quotes / NBBO inputs	Massive US Stocks SIP - quotes_v1	Later
Nasdaq HALTs / resumptions	Existing Nasdaq source and QuantLab HALT engine	Primary specialized source
SEC filings / regulatory data	SEC EDGAR	Primary specialized source
Venue-specific microstructure / order book	Direct venue feeds / Databento if later required	Future optional

4. Why Massive/Polygon is the MVP fit

- Matches the required consolidated U.S. market view rather than a Nasdaq-only venue view.
- Provides ready-made 1-minute OHLCV for all U.S. equities as daily downloadable flat files.
- Fits QuantLab's immutable RAW archive pattern and avoids thousands of per-symbol API calls.
- Allows a later validation path using tick trades and quotes from the same provider.
- Keeps the Replay Engine independent from the provider through the Source Adapter and canonical model.
- Provides long historical coverage; Massive currently documents minute aggregates back to September 10, 2003, subject to plan access.

5. Updated target architecture

```

Massive / Polygon - US Stocks SIP
|
+-- minute_aggs_v1  [MVP]
+-- trades_v1       [validation / future]
+-- quotes_v1       [future]
|
v
MassiveSourceAdapter
|
v
Immutable RAW Archive
|
v
Data Quality / Validation
|
v
PostgreSQL Canonical Model
  core.market_bar_1m
  market_scope = CONSOLIDATED_US
  |
  v
Historical Data API
  |
  v
Replay Engine + Simulated Clock
  |
  v
MarketContext(as_of=T)
  |
  v
Aggregation -> Features -> Models -> Evaluation

```

6. Canonical market data model

```
core.market_bar_1m
  security_id
  timestamp
  session
  open
  high
  low
  close
  volume
  transaction_count (when available)
  market_scope = CONSOLIDATED_US
  source_provider = MASSIVE
  source_dataset = us_stocks_sip/minute_aggs_v1
  dataset_version
  ingestion_run_id
```

The canonical table must not embed Massive-specific business logic beyond provenance metadata. Replay, features and models consume QuantLab canonical data, never the provider API directly.

7. Raw archive and normalization

```
Provider flat file
  ↓
RAW / immutable / checksum
  ↓
Parser + normalization
  ↓
Data-quality controls
  ↓
Canonical PostgreSQL rows
```

Massive documents that stock Flat Files are unadjusted. QuantLab must therefore preserve the original values and handle corporate-action adjustment as an explicit, versioned transformation when a backtest requires adjusted history.

8. Validation strategy with Massive

The validation design has two complementary levels. First, `minute_aggs_v1` is validated for integrity, continuity, timestamps, OHLC consistency and comparison against independent reference days. Second, selected Golden Market Days can be reconstructed from `trades_v1` to validate QuantLab aggregation logic against Massive minute aggregates, while documenting trade eligibility/sale-condition rules when differences arise.

```
Massive trades_v1
  ↓
QuantLab aggregation rules
  ↓
Calculated 1M OHLCV
  |
  ├── compare — Massive minute_aggs_v1
  |
  └── compare — independent visual/reference source
```

9. Architecture options - revised

Option	Description	Position after provider decision
A - PostgreSQL-centric	Massive flat files -> RAW -> PostgreSQL -> Replay	RECOMMENDED MVP
B - PostgreSQL + Parquet	Add canonical Parquet for large multi-year	Recommended evolution when scale justifies

	batch analytics	it
C - Event/tick architecture	Use trades/quotes/direct feeds for event-by-event replay	Future specialized research
D - External backtesting engine	QuantLab data into LEAN or another execution engine	Potential Phase 2 complement

10. Option A - MVP baseline

Massive minute_aggs_v1 -> RAW archive -> PostgreSQL -> Replay -> Features -> Predictive Model -> Viewer/Batch

This is the shortest path to the functional requirement represented in the reference diagram: replay a ticker/day from a selected time, render candles and volume, calculate indicators at each simulated minute and evaluate model predictions without look-ahead.

11. Option B - scalable analytical evolution

Massive flat files -> RAW -> Canonical Parquet + PostgreSQL metadata/results -> Batch + Replay

Parquet becomes attractive when the universe and history make repeated PostgreSQL scans inefficient. This evolution does not change the Replay Engine interfaces if the Historical Data API remains stable.

12. Option C - tick / microstructure extension

Massive trades_v1 / quotes_v1 or direct venue feeds -> event store -> event replay -> dynamic bars/features

This is not required for the MVP. It becomes useful for spread, bid/ask, liquidity, trade-condition research and later execution realism. Direct venue feeds can be added if consolidated SIP data is insufficient for a specific research question.

13. Option D - external execution engine

LEAN or another external trading engine remains a possible downstream consumer for Phase 2 execution simulation. It does not replace QuantLab's scientific Replay Engine, point-in-time MarketContext or proprietary Feature Engine.

14. Updated development baseline

FIRST VERTICAL SLICE

```

Massive minute_aggs_v1
↓
ONE TICKER + ONE DAY
↓
RAW archive + PostgreSQL
↓
Replay Clock
↓
1M + 5M aggregation
↓
VWAP + HOD/LOD + Volume
↓
Interactive Viewer
↓
PLAY minute-by-minute
↓
Independent validation

```

15. Repository artifacts

```
docs/architecture/diagrams/
  QuantLab_US_Equities_Functional_Architecture.drawio
  QuantLab_US_Equities_Functional_Architecture.svg    [export for Markdown]

docs/architecture.md
docs/database.md
docs/data_sources.md
```

The .drawio file is the editable source of truth for the functional diagram. The supplied reference PDF was used as the visual/topological baseline for this revision. The SVG export should be regenerated from draw.io whenever the source diagram changes.

16. Decision summary

Decision	Baseline v2
Platform scope	US Equities Historical Replay
Primary market provider	Massive (formerly Polygon.io)
Primary market dataset	US Stocks SIP minute_aggs_v1
Market scope	CONSOLIDATED_US - CTA A/B + UTP C consolidated view
MVP granularity	OHLCV 1 minute
Raw retention	Original Massive flat files + checksum/provenance
Operational store	PostgreSQL
Future analytical store	Canonical Parquet if/when scale requires
Nasdaq events	Existing Nasdaq HALT engine
SEC	SEC EDGAR
Trades/quotes	Massive optional validation/future research
Venue microstructure	Optional future direct feeds / Databento
Replay	QuantLab custom engine
Viewer + Batch	Same Replay Engine